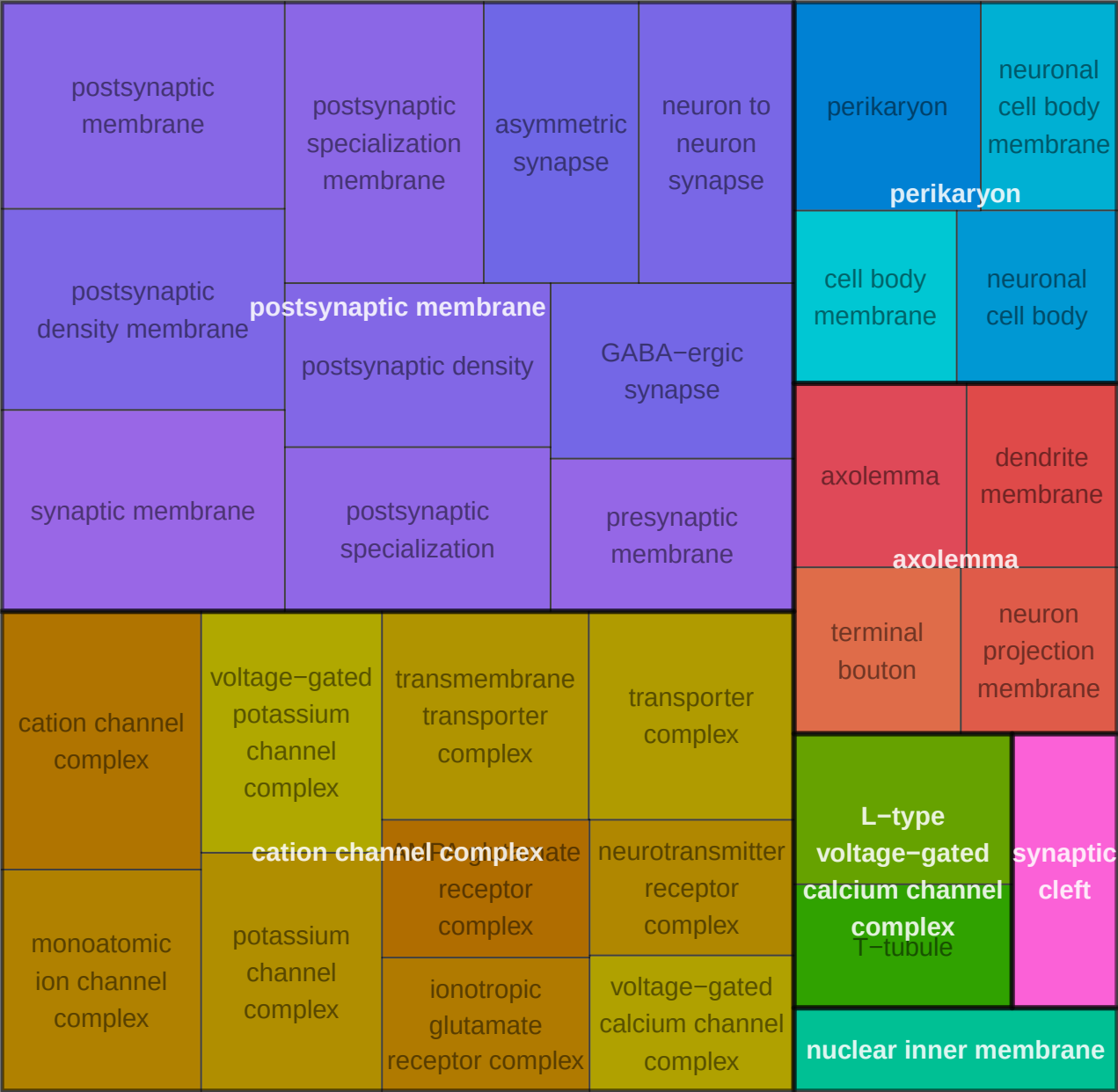


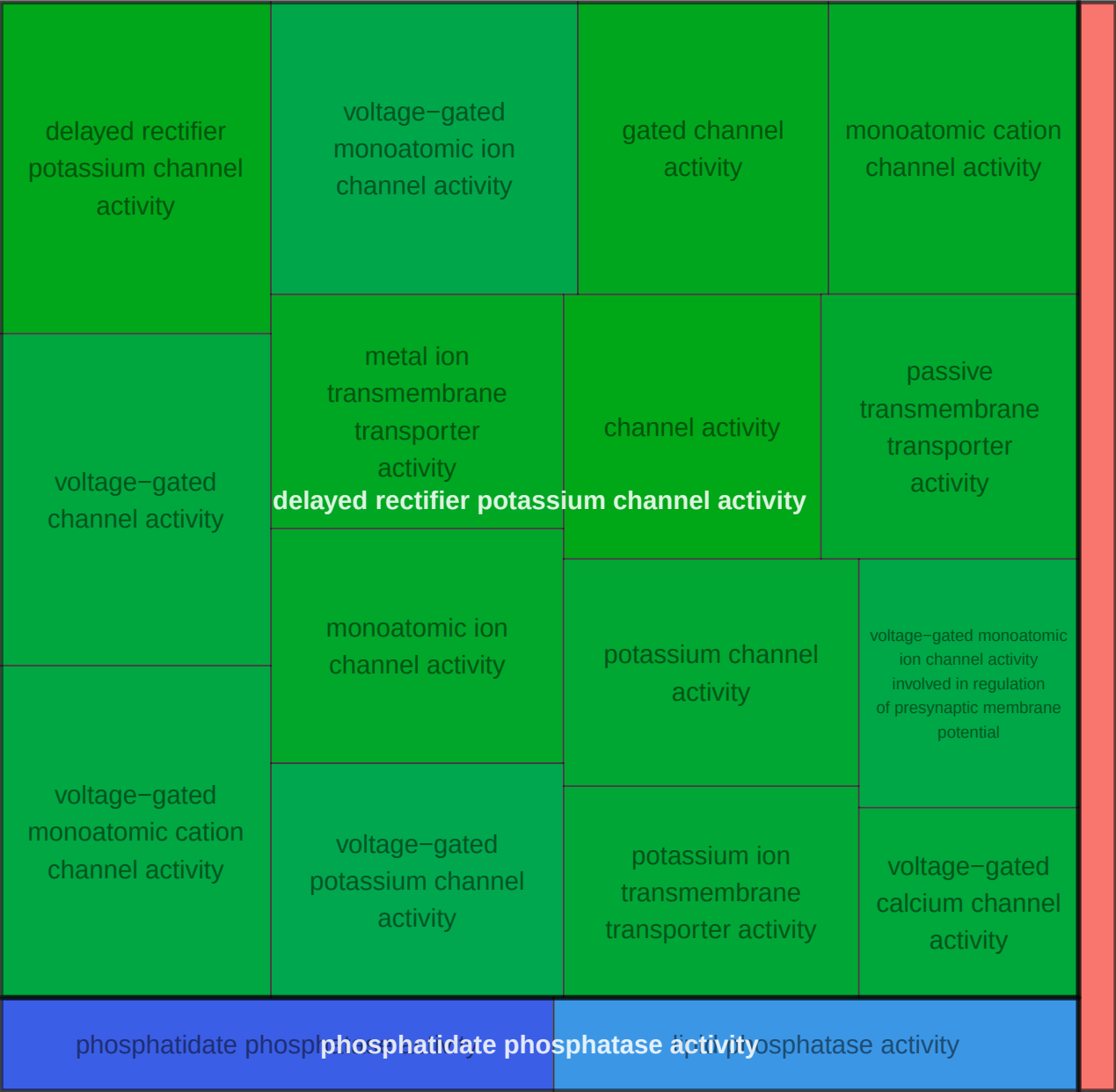
Astro-1_0_upEA_UpAA CC



Astro-1_0_upEA_UpAA BP

modulation of chemical synaptic transmission	regulation of synaptic transmission, glutamatergic	cellular response to ammonium ion	cellular response to nitric oxide	response to ammonium ion
synaptic transmission, glutamatergic				
regulation of trans-synaptic signaling	synaptic transmission, glutamatergic	potassium ion transport	potassium ion transport	membrane hyperpolarization
neurotransmitter receptor diffusion trapping	neurotransmitter receptor transport, postsynaptic endosome to lysosome			
neurotransmitter receptor transport, postsynaptic endosome to lysosome				
postsynaptic neurotransmitter receptor diffusion trapping	receptor diffusion trapping	nitric oxide-cGMP-mediated signaling	regulation of AMPA receptor activity	response to light intensity

Astro-1_0_upEA_UpAA MF



Astro-1_AA_down BP

nuclear-transcribed mRNA poly(A) tail shortening	piRNA processing	nuclear-transcribed mRNA catabolic process, nonsense-mediated decay	mRNA destabilization	blastocyst formation	dorsal/ventral neural tube patterning	in utero embryonic development
nuclear-transcribed mRNA catabolic process, deadenylation-dependent decay	RNA 3'-end processing	RNA destabilization	nuclear-transcribed mRNA catabolic process	dorsal/ventral neural tube patterning morphogenesis of embryonic epithelium		
positive regulation of mRNA catabolic process	positive regulation of mRNA metabolic process	regulation of mRNA stability	regulation of mRNA catabolic process	blastocyst development	neural tube development	
regulatory ncRNA processing	regulatory ncRNA-mediated gene silencing	regulation of RNA stability	mRNA catabolic process	neural tube formation	embryonic epithelial tube formation	protein localization to ciliary transition zone
cell differentiation in spinal cord	spinal cord motor neuron differentiation	central nervous system neuron development	spinal cord development	neural tube formation		
	central nervous system neuron axonogenesis	central nervous system neuron differentiation	telencephalon development	epithelial tube formation	tube formation	protein localization to cilium
central nervous system neuron axonogenesis	ventral spinal cord development	central nervous system neuron differentiation	telencephalon development	regulation of smoothed signaling pathway	dorsal/ventral pattern formation	

Astro-1_AA_up CC

postsynaptic density membrane		postsynaptic specialization membrane		asymmetric synapse		postsynaptic density		perikaryon					
synaptic membrane		postsynaptic specialization		neuron to neuron synapse		presynaptic endocytic zone membrane		dendritic perikaryon					
						photoreceptor ribbon synapse		neuronal cell body					
postsynaptic membrane		presynaptic membrane		Schaffer collateral – CA1 synapse		presynaptic endocytic zone		neuronal cell body membrane					
				GABA-ergic synapse		ribbon synapse		endocytic vesicle					
cation channel complex		AMPA glutamate receptor complex		ionotropic glutamate receptor complex		voltage-gated calcium channel complex		calcium channel complex		voltage-gated potassium channel complex		endocytic vesicle	
												endocytic vesicle membrane	
monoatomic ion channel complex		transmembrane transporter complex		transporter complex		neurotransmitter receptor complex		potassium channel complex		L-type voltage-gated calcium channel complex		axolemma	
								plasma membrane signaling receptor complex				synaptic cleft	

Astro-1_AA_up BP



Astro-1_AA_up MF

voltage-gated channel activity		voltage-gated monoatomic cation channel activity		voltage-gated monoatomic ion channel activity		metal ion transmembrane transporter activity		phosphatidylinositol bisphosphate binding
gated channel activity		monoatomic ion channel activity		voltage-gated potassium channel activity		potassium channel activity		proline:sodium symporter activity
monoatomic cation channel activity		voltage-gated channel activity		calcium channel activity		calcium ion transmembrane transporter activity		neutral L-amino acid:sodium symporter activity
delayed rectifier potassium channel activity		passive transmembrane transporter activity		potassium ion transmembrane transporter activity		voltage-gated monoatomic ion channel activity involved in regulation of presynaptic membrane potential		amino acid:sodium symporter activity
AP-1 adaptor complex binding		voltage-gated calcium channel activity		branched-chain amino acid transmembrane transporter activity		neurotransmitter transmembrane transporter activity		amino acid:monoatomic cation symporter activity
AP-1 adaptor complex binding								

Macro_AA_up CC



Macro_AA_up BP



Macro_AA_up MF

dopamine receptor binding

beta-catenin binding

chromatin DNA binding

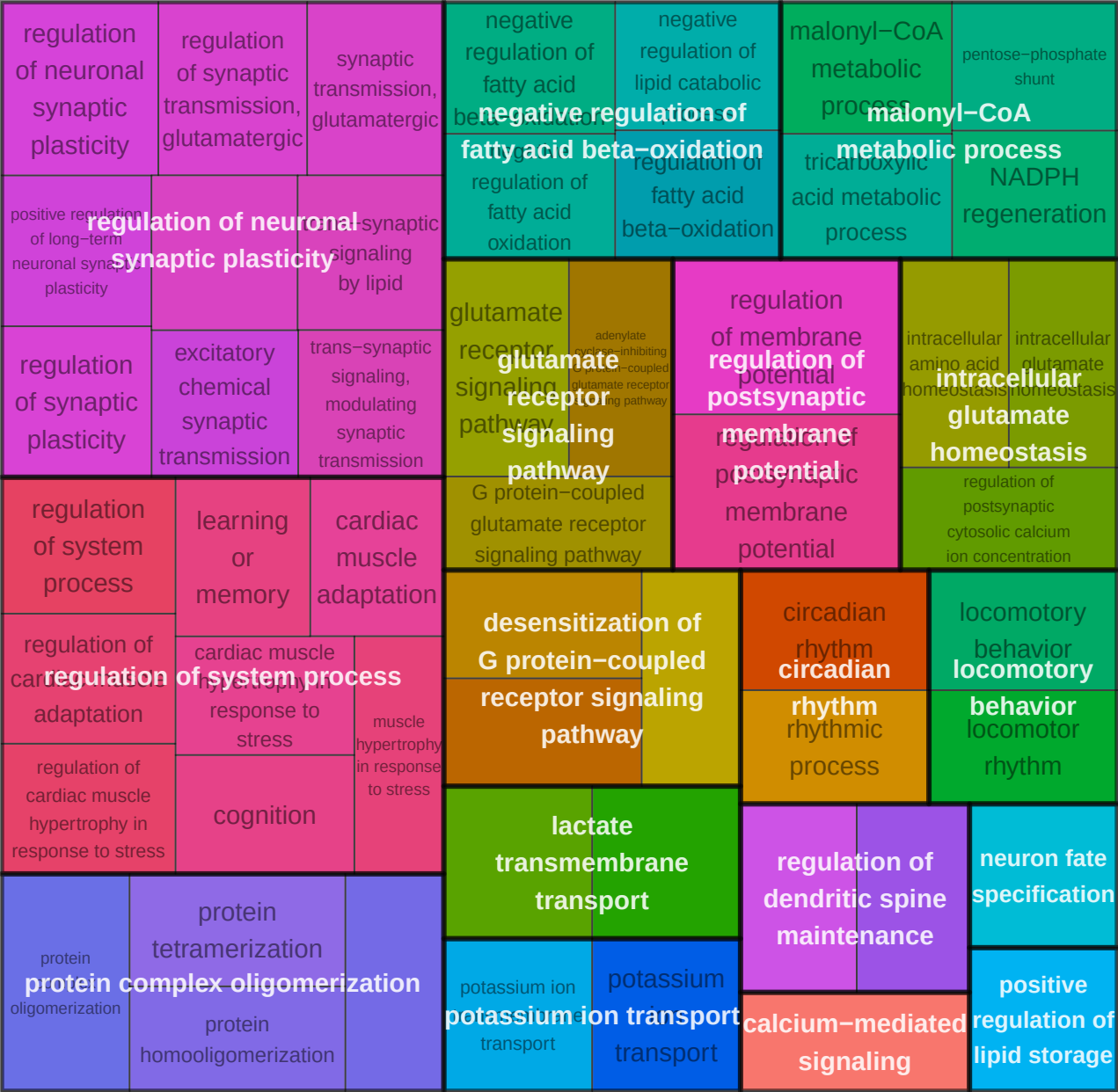
G-protein
beta/gamma-subunit
complex binding

transcription coregulator binding

Micro-1_0_upEA_UpAA CC



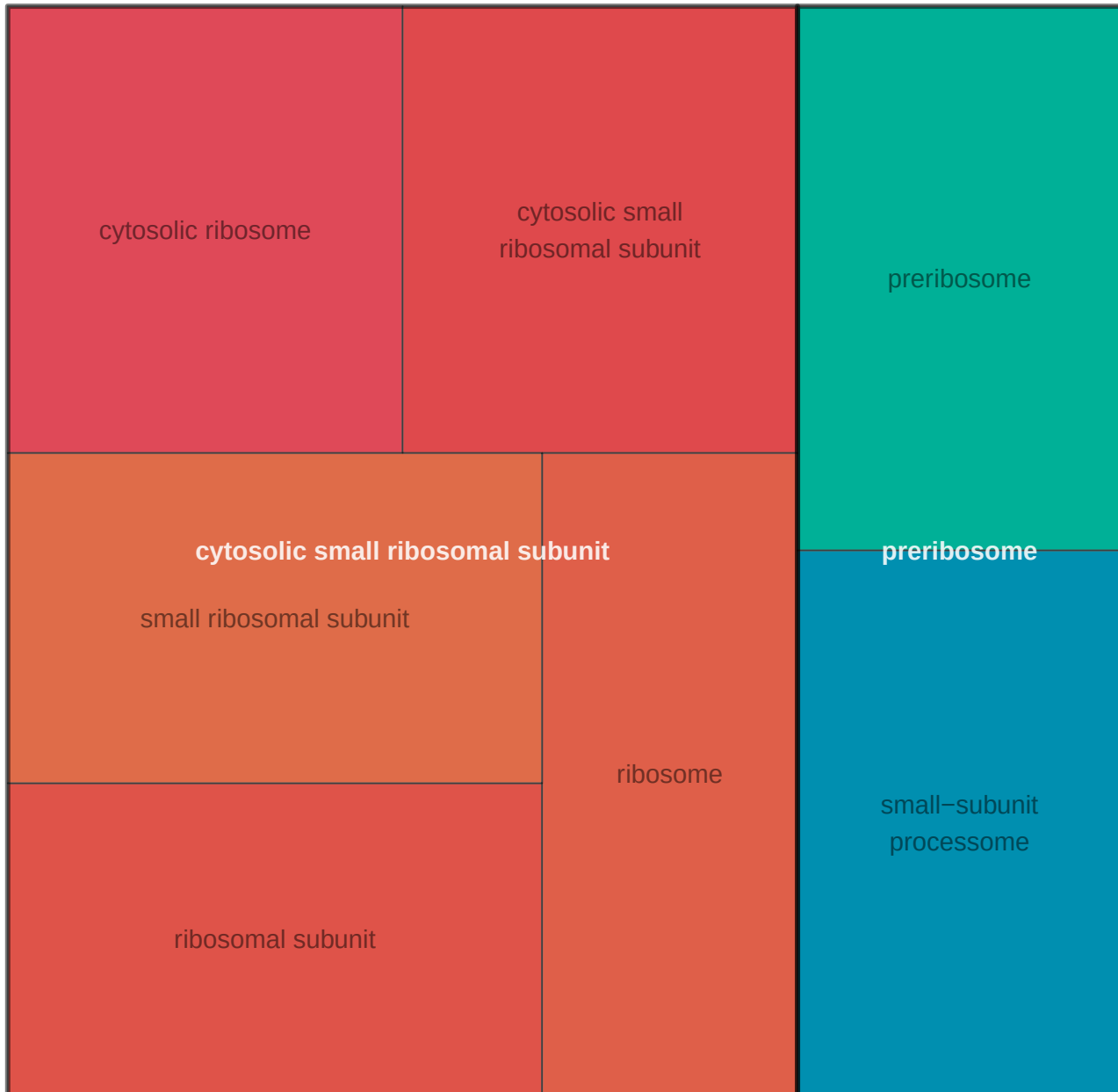
Micro-1_0_upEA_UpAA BP



Micro-1_0_upEA_UpAA MF



Micro-1_AA_down CC



Micro-1_AA_down BP

negative regulation of post-translational protein modification	negative regulation of protein modification by small protein conjugation or removal	negative regulation of protein ubiquitination	neural tube closure	primary neural tube formation	positive regulation of intrinsic apoptotic signaling pathway by p53 class mediator	positive regulation of signal transduction by p53 class mediator
negative regulation of proteolysis involved in protein catabolic process	cytoplasmic translation	regulation of ubiquitin-dependent protein catabolic process	tube closure	embryonic epithelial tube formation	positive regulation of intrinsic apoptotic signaling pathway by p53 class mediator	
negative regulation of ubiquitin-dependent protein catabolic process	regulation of protein ubiquitination	regulation of protein modification by small protein conjugation or removal	epithelial tube formation	tube formation	regulation of signal transduction by p53 class mediator	signal transduction by p53 class mediator
negative regulation of proteolysis	regulation of post-translational protein modification	regulation of proteolysis involved in protein catabolic process	neural tube formation			
negative regulation of molecular function	regulation of ubiquitin protein ligase activity	regulation of ubiquitin-protein transferase activity	morphogenesis of embryonic epithelium	mesenchymal cell differentiation	neural crest cell differentiation	
negative regulation of protein ligase activity	negative regulation of NF-kappaB transcription factor activity	negative regulation of transferase activity	neural tube development	mesenchyme development	rRNA processing	rRNA processing
negative regulation of ubiquitin-protein transferase activity	negative regulation of DNA-binding transcription factor activity	negative regulation of catalytic activity	positive regulation of intrinsic apoptotic signaling pathway	regulation of intrinsic apoptotic signaling pathway	ribosomal small subunit biogenesis	protein stabilization

Micro-1_AA_down MF

ubiquitin ligase inhibitor activity

ubiquitin-protein transferase regulator activity

ubiquitin-protein
transferase
inhibitor activity

ubiquitin-protein
transferase
regulator activity

NF-kappaB binding

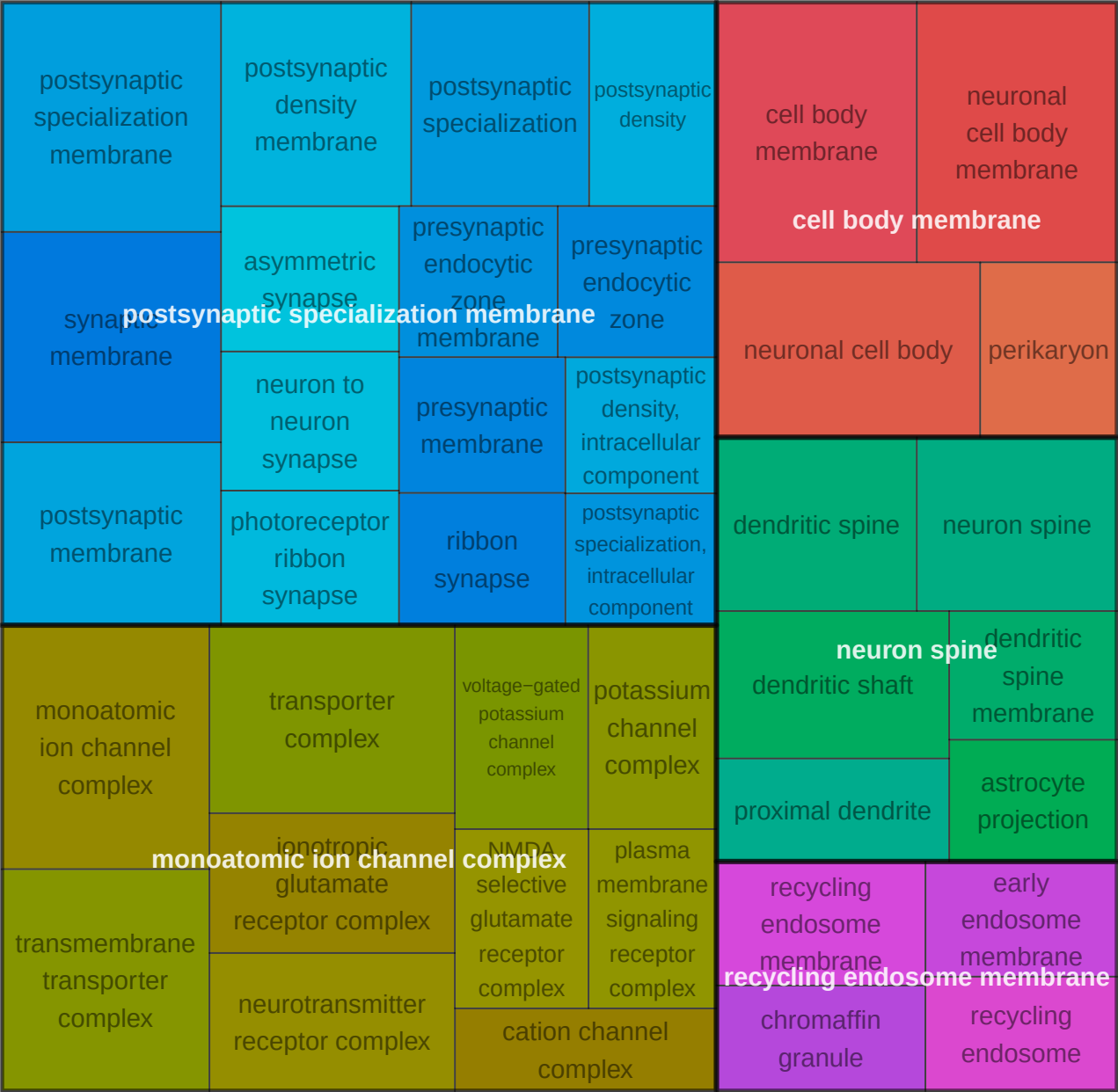
mRNA 5'-UTR binding

mRNA 5'-UTR binding

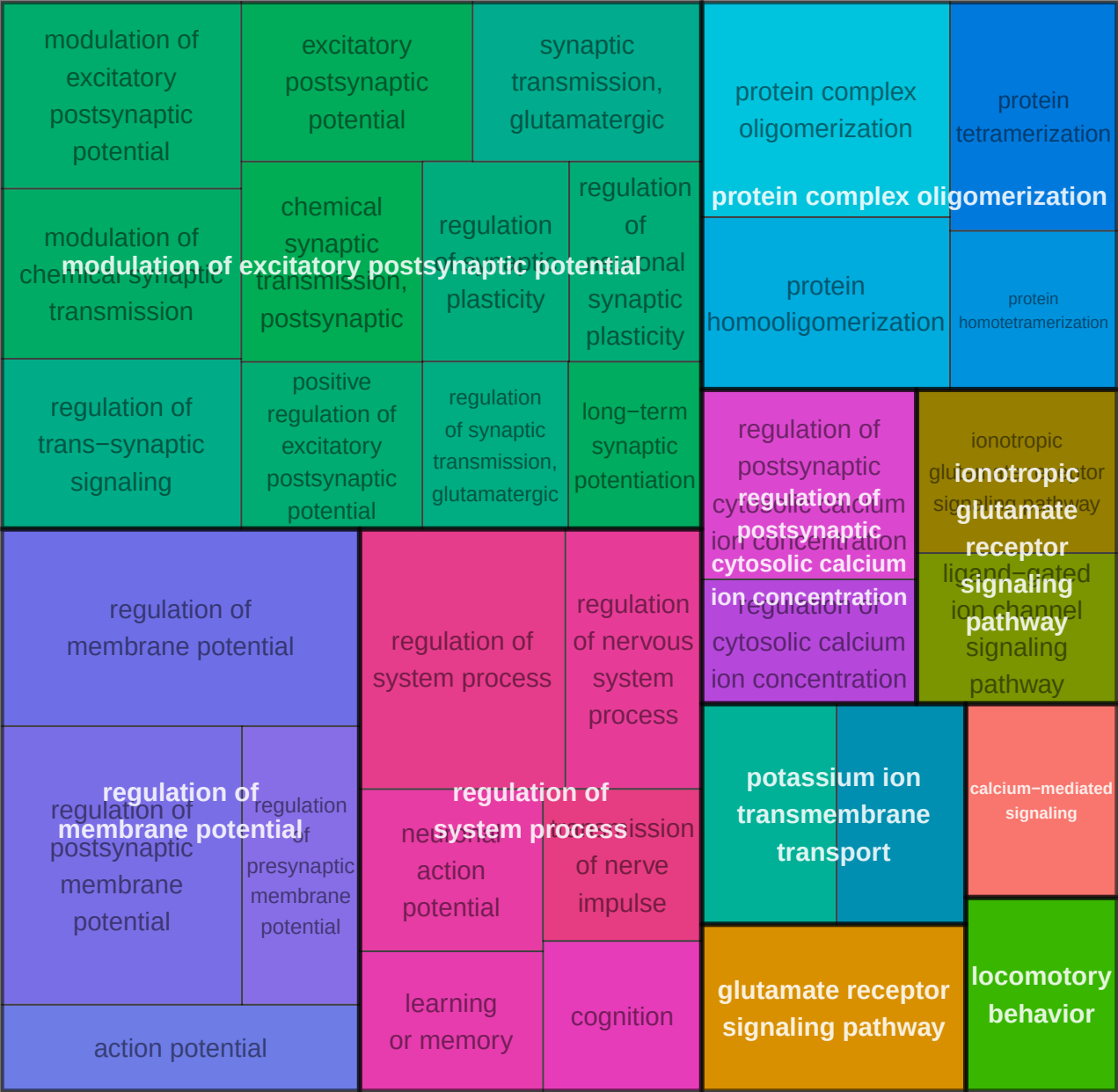
mRNA 3'-UTR binding

**RNA
methyltransferase
activity**

Micro-1_AA_up CC



Micro-1_AA_up BP



Micro-1_AA_up MF

gated channel activity	metal ion transmembrane transporter activity	monoatomic ion channel activity	channel activity		passive transmembrane transporter activity		adenosine receptor binding		beta-2 adrenergic receptor binding		
	glutamate-gated calcium ion channel activity		ligand-gated monoatomic cation channel activity		voltage-gated monoatomic cation channel activity		potassium ion transmembrane transporter activity		adenosine receptor binding		
monoatomic cation channel activity	voltage-gated potassium channel activity	gated channel activity	ligand-gated calcium channel activity		extracellular ligand-gated monoatomic ion channel activity	amino acid transmembrane transporter activity	calcium channel activity		peptide hormone receptor binding	hormone receptor binding	
voltage-gated channel activity			ligand-gated channel activity		calcium ion transmembrane transporter activity	organic acid transmembrane transporter activity	organic anion transmembrane transporter activity	carboxylic acid binding		organic acid binding	
voltage-gated monoatomic ion channel activity	ligand-gated monoatomic ion channel activity involved in regulation of presynaptic membrane potential		ligand-gated channel activity					glutamate binding		glycine binding	
	potassium channel activity		ligand-gated monoatomic ion channel activity		carboxylic acid transmembrane transporter activity	delayed rectifier potassium channel activity	outward rectifier potassium channel activity	amide binding		amyloid-beta binding	
glutamate receptor activity		neurotransmitter receptor activity	glutamate receptor activity	transmitter-gated channel activity	AMPA glutamate receptor activity	G protein-coupled receptor activity involved in regulation of postsynaptic membrane potential		CoA carboxylase activity		adenylate cyclase binding	
postsynaptic neurotransmitter receptor activity	glutamate-gated receptor activity	neurotransmitter receptor activity involved in regulation of postsynaptic membrane potential		transmitter-gated monoatomic ion channel activity	NMDA glutamate receptor activity				immunoglobulin binding	mRNA regulatory element binding repressor activity	
					adenylate cyclase inhibiting G protein-coupled glutamate receptor activity				myosin V binding	G-protein beta-subunit binding	

Micro-3_EA_up CC



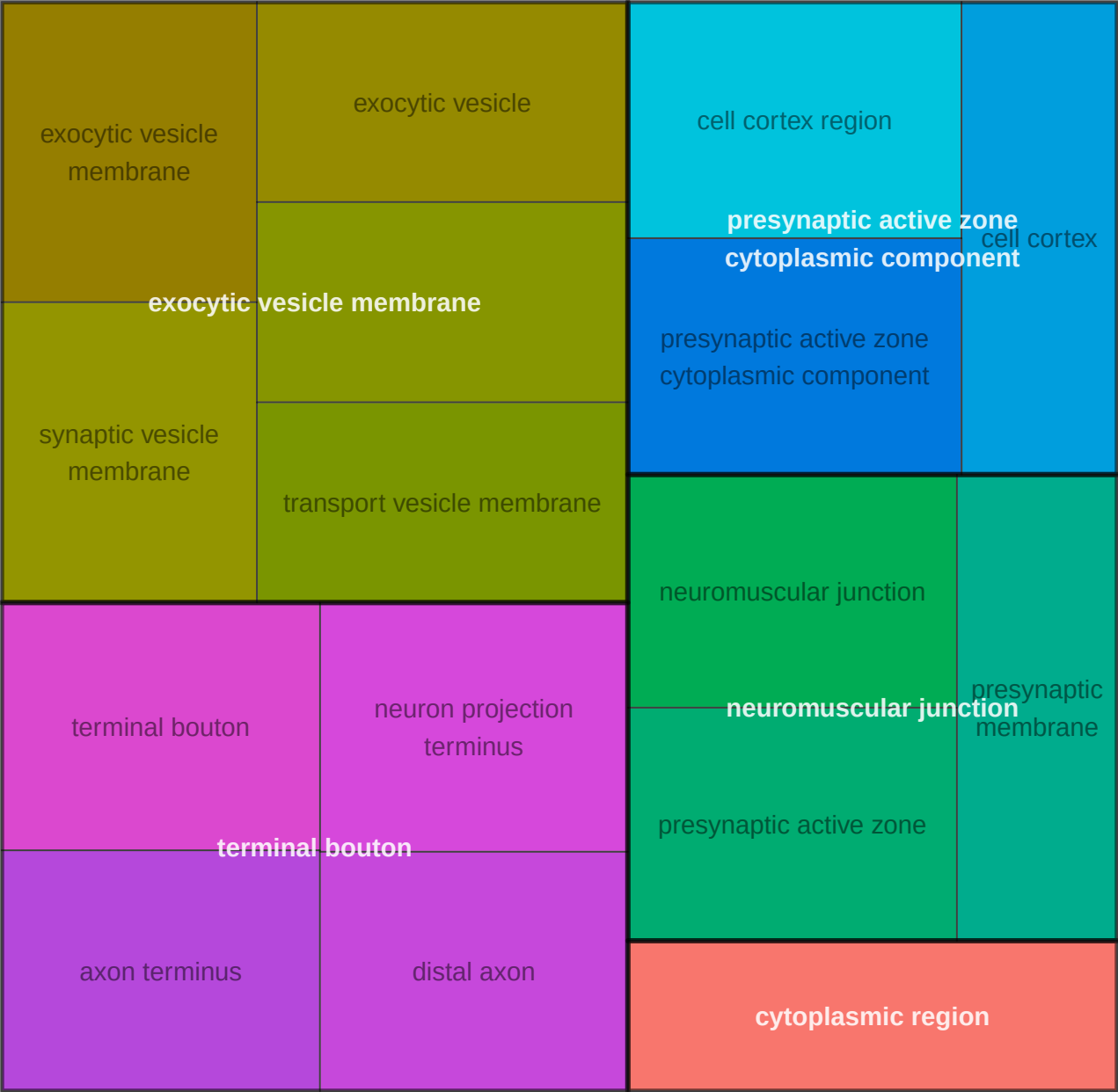
Micro-3_EA_up BP



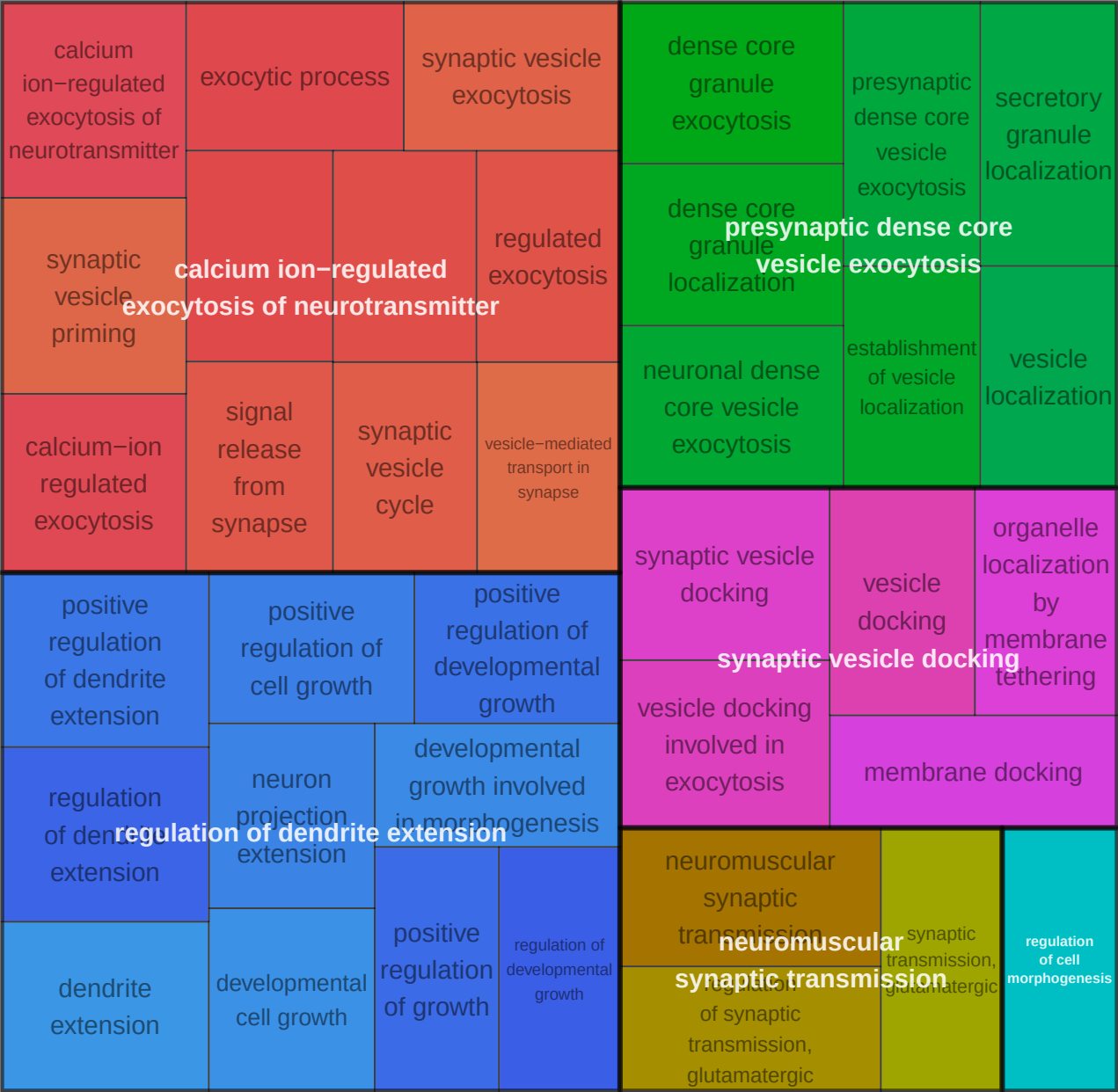
Micro-3_EA_up MF

GABA-gated chloride ion channel activity		extracellular ligand-gated monoatomic ion channel activity		chloride transmembrane transporter activity		monoatomic anion transmembrane transporter activity		adenylate cyclase regulator activity	
ligand-gated monoatomic ion channel activity involved in regulation of presynaptic membrane potential		ligand-gated monoatomic ion channel activity involved in regulation of presynaptic membrane potential		inorganic anion transmembrane transporter activity		ligand-gated monoatomic ion channel activity		cyclase regulator activity	
ligand-gated monoatomic anion channel activity				ligand-gated channel activity				cyclase regulator activity	
G protein-coupled neurotransmitter receptor activity		GABA-A receptor activity		neurotransmitter receptor activity involved in regulation of postsynaptic membrane potential		transmitter-gated channel activity		GABA receptor binding	
G protein-coupled receptor activity		GABA receptor activity		transmitter-gated monoatomic ion channel activity involved in regulation of postsynaptic membrane potential		postsynaptic neurotransmitter receptor activity		G-protein alpha-subunit binding	
				transmitter-gated monoatomic ion channel activity		neurotransmitter receptor activity			

Oligo-1_0_upEA_UpAA CC



Oligo-1_0_upEA_UpAA BP



Oligo-1_0_upEA_UpAA MF

syntaxin-1 binding

syntaxin-1 binding

syntaxin binding

SNARE binding

**calmodulin
binding**

diacylglycerol binding

Oligo-1_AA_up CC



Oligo-1_AA_up MF

P-type calcium
transporter activity

P-type ion
transporter
activity

calmodulin binding

P-type calcium transporter activity

P-type transmembrane
transporter activity

ATPase-coupled
monoatomic cation
transmembrane
transporter activity

glutamate receptor binding

calcium/calmodulin-dependent
protein kinase activity

syntaxin-1 binding

syntaxin-1 binding

diacylglycerol binding

T-tubule

T-tubule

sarcolemma

intercalated disc

Olgo-3_0_downEA_UpAA CC

alpha-ketoacid
dehydrogenase complex

oxoglutarate dehydrogenase complex

tricarboxylic acid cycle heteromeric enzyme complex

SAGA complex

gap junction

tricarboxylic acid cycle
heteromeric enzyme complex

SAGA-type complex

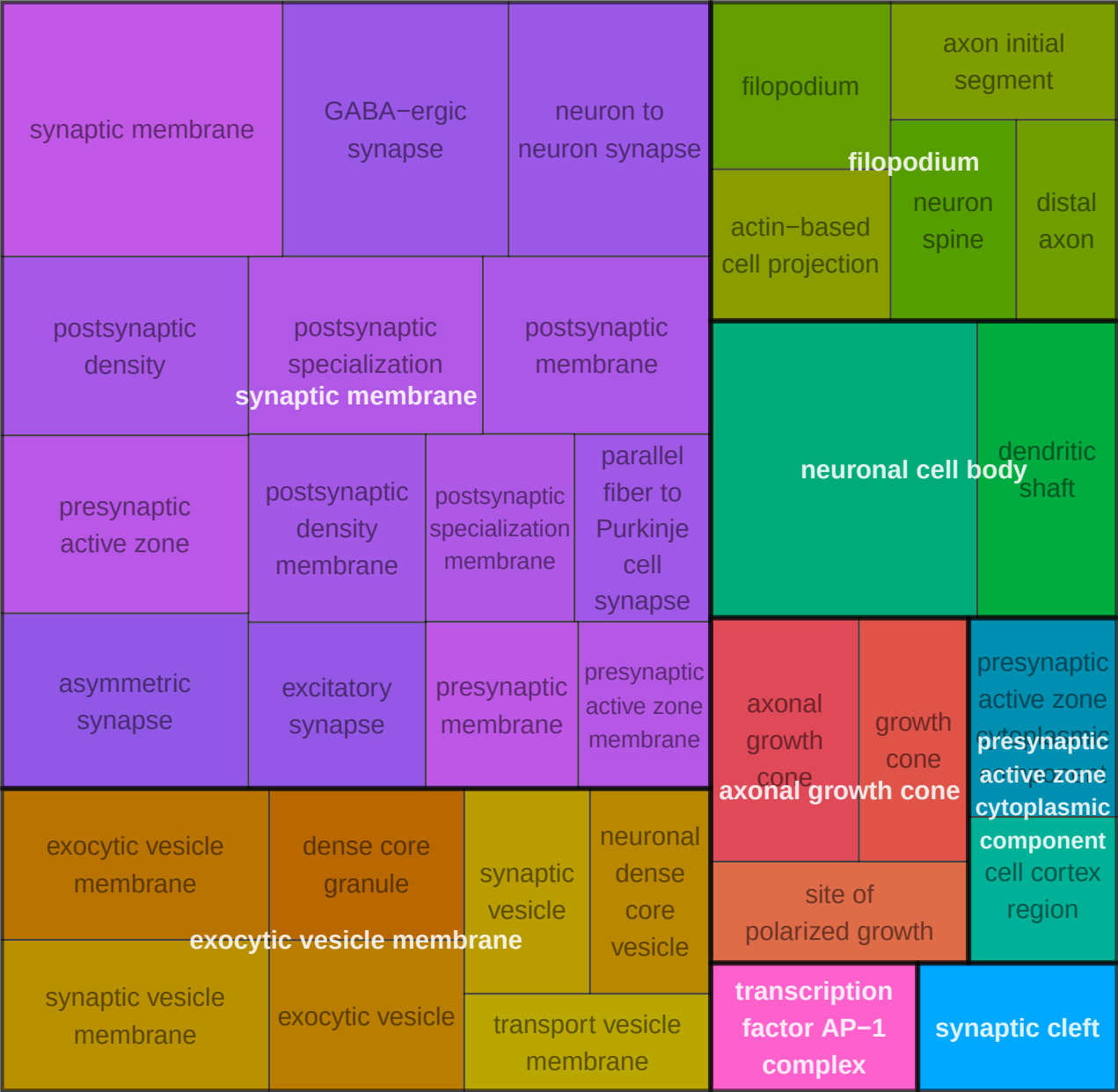
Oligo-3_0_downEA_UpAA BP



Oligo-3_0_downEA_UpAA MF

histone H2AK5 acetyltransferase activity	histone H2AK9 acetyltransferase activity	histone H2BK12 acetyltransferase activity	histone H2BK5 acetyltransferase activity	histone H3K18 acetyltransferase activity	hydro-lyase activity		tRNA (guanine) methyltransferase activity
histone H3K23 acetyltransferase activity	histone H3K56 acetyltransferase activity	histone H3K9 acetyltransferase activity	histone H4K16 acetyltransferase activity	histone H2B acetyltransferase activity	hydro-lyase activity		tRNA (guanine) methyltransferase activity
histone H3K27 acetyltransferase activity	histone H2A acetyltransferase activity	histone H4K12 acetyltransferase activity	histone H4K5 acetyltransferase activity	histone H4K8 acetyltransferase activity	lyase activity		RNA methyltransferase activity
histone H3K36 acetyltransferase activity	histone H3 acetyltransferase activity	histone H4 acetyltransferase activity	N-acetyltransferase activity		S-adenosyl-L-methionine binding	heme binding	
histone H3K4 acetyltransferase activity	histone H3K14 acetyltransferase activity	histone acetyltransferase activity	protein N-acetyltransferase activity	N-acyltransferase activity	S-adenosyl-L-methionine binding	heme binding	
				acetyltransferase activity	sulfur compound binding	tetrapyrrole binding	
nitrite reductase (NO-forming) oxidoreductase activity, acting on other nitrogenous compounds as donors, cytochrome as acceptor		nitrite reductase activity		pyridoxal phosphate binding		nitric oxide binding	ubiquitin protein ligase binding
oxidoreductase activity, acting on other nitrogenous compounds as donors, cytochrome as acceptor		oxidoreductase activity, acting on other nitrogenous compounds as donors		pyridoxal phosphate binding vitamin B6 binding			ubiquitin-like protein ligase binding
				pyridoxal vitamin B6 binding		oxygen binding	modified amino acid binding
							protein folding chaperone

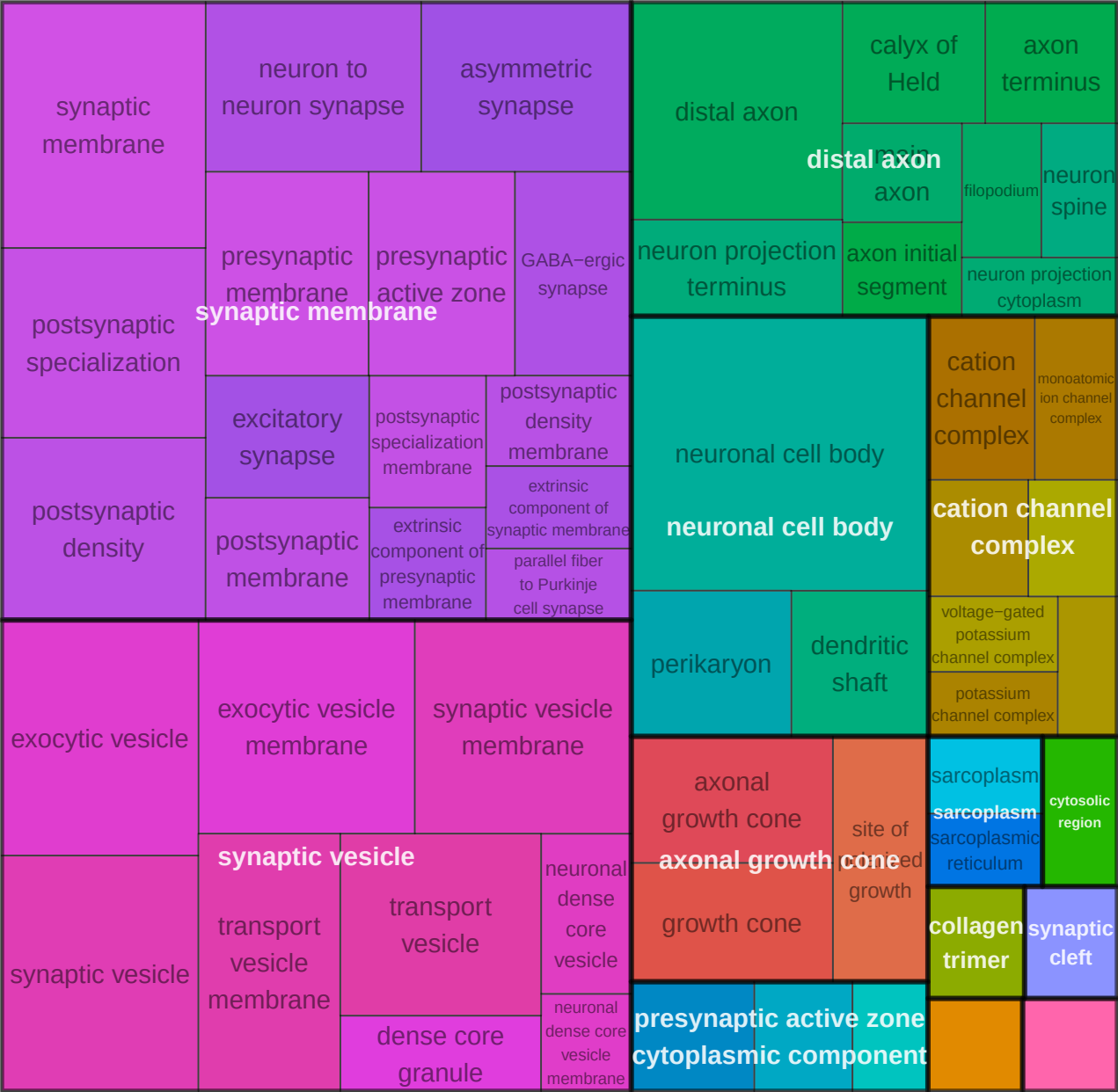
Oligo-3_0_upEA_UpAA CC



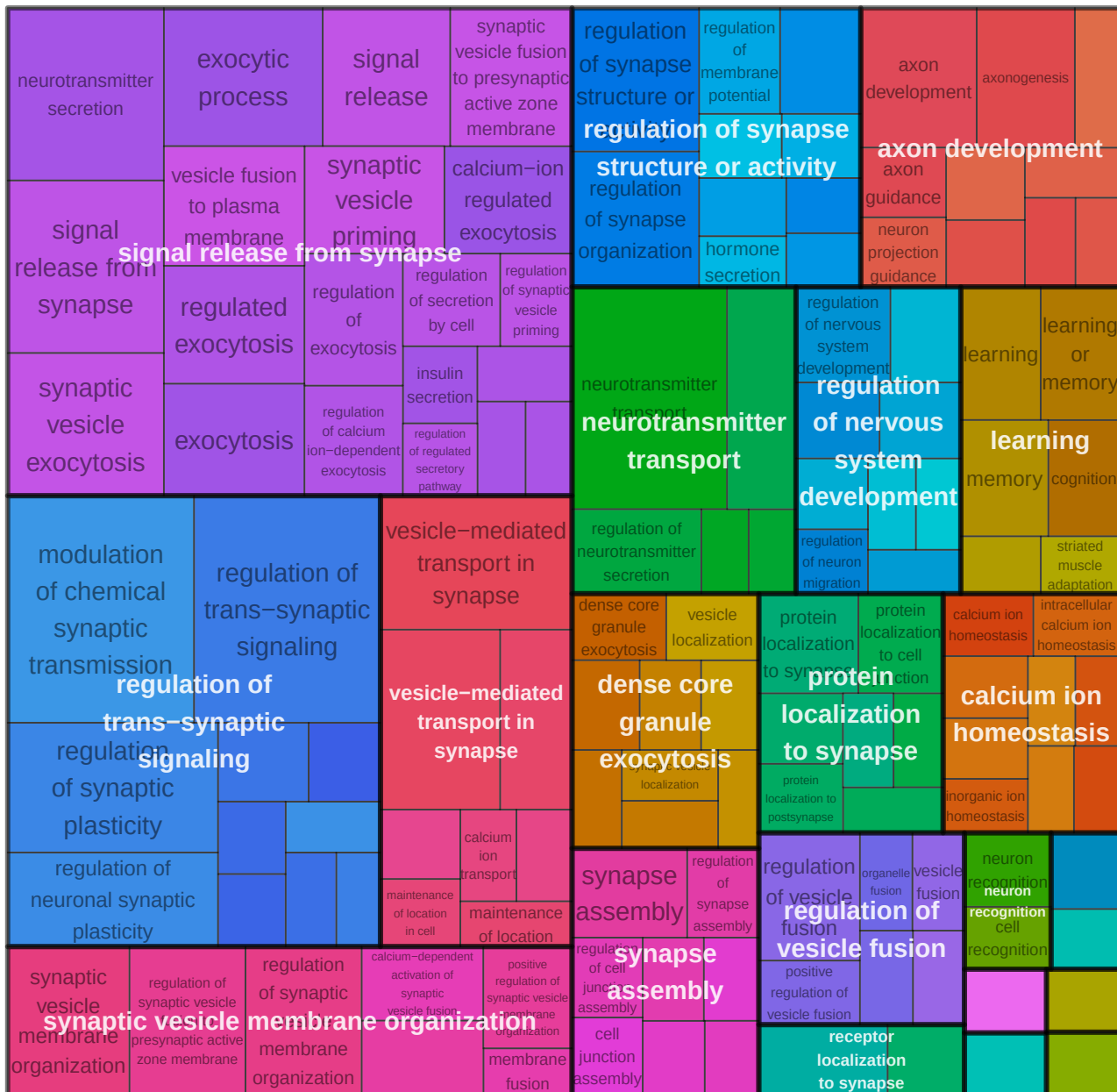
Oligo-3_0_upEA_UpAA BP



Oligo-3_AA_up CC



Oligo-3_AA_up BP



Oligo-3_AA_up MF



Oligo-3_EA_up CC



Oligo-3_EA_up BP



Oligo-3_EA_up MF

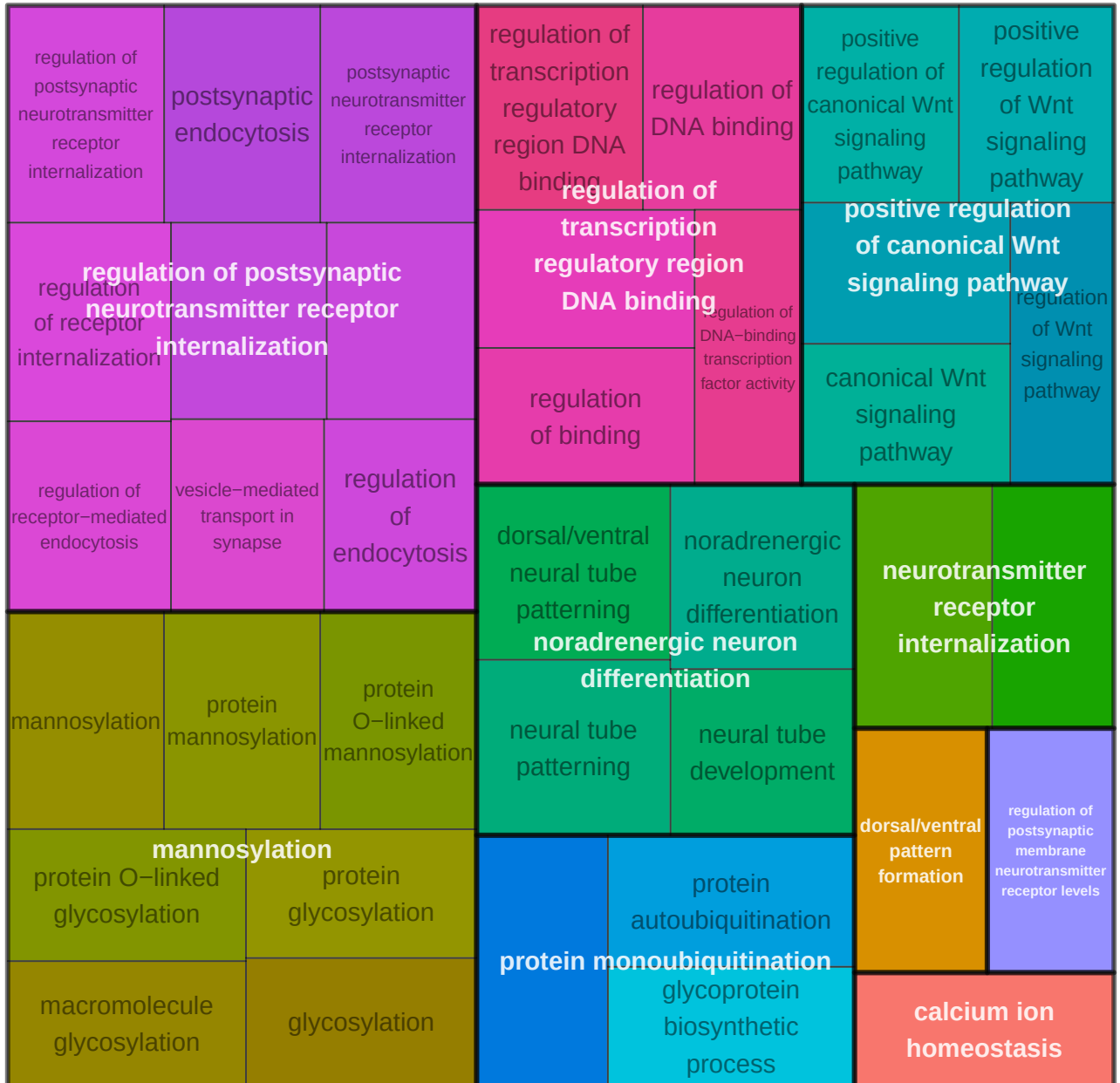
calcium/calmodulin-dependent
protein kinase activity

glycosaminoglycan binding

extracellular matrix structural constituent

growth factor binding

OPC-4_AA_down BP



OPC-4_AA_down MF

dolichyl-phosphate-mannose-protein
mannosyltransferase activity

**dolichyl-phosphate-mannose-protein
mannosyltransferase activity**

mannosyltransferase activity

beta-catenin binding

Vasc-Endo_EA_up CC



Vasc-VLMC_AA_down CC

filopodium tip

filopodium tip

filopodium

actin-based
cell projection

**nuclear
membrane**

perinuclear endoplasmic reticulum

Vasc-VLMC_AA_down BP

